MA 311 - Formula Sheet

Derivatives

$$\frac{d}{dx}\left(a^x\right) = \ln a \cdot a^x$$

$$\frac{d}{dx}\left(\log_a x\right) = \frac{1}{\ln a} \cdot \frac{1}{x}$$

$$\frac{d}{dx}\left(\sin x\right) = \cos x$$

$$\frac{d}{dx}\left(\cos x\right) = -\sin x$$

$$\frac{d}{dx}\left(\csc x\right) = -\csc x \cot x$$

$$\frac{d}{dx}\left(\sec x\right) = \sec x \tan x$$

$$\frac{d}{dx}\left(\tan x\right) = \sec^2 x$$

$$\frac{d}{dx}\left(\cot x\right) = -\csc^2 x$$

$$\frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\left(\cos^{-1} x\right) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\left(\csc^{-1} x\right) = \frac{-1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}\left(\sec^{-1} x\right) = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}\left(\tan^{-1} x\right) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}\left(\cot^{-1} x\right) = \frac{-1}{1+x^2}$$

Anti-derivatives

$$\int \cos x \; dx = \sin x + C$$

$$\int \sin x \; dx = -\cos x + C$$

$$\int \tan x \; dx = -\ln|\cos x| + C$$

$$\int \cot x \; dx = \ln|\sin x| + C$$

$$\int \csc x \; dx = -\ln|\csc x + \cot x| + C$$

$$\int \sec x \; dx = \ln|\sec x + \tan x| + C$$

$$\int \csc^2 x \; dx = -\cot x + C$$

$$\int \sec^2 x \; dx = \tan x + C$$

$$\int \csc x \cot x \; dx = -\csc x + C$$

$$\int \sec x \tan x \; dx = \sec x + C$$

$$\int \frac{1}{\sqrt{a^2-x^2}} \; dx = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{x^2+a^2} \; dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$$

$$\int a^x \; dx = \frac{1}{\ln a} \cdot a^x + C$$

$$\int \frac{1}{x\sqrt{x^2-a^2}} \; dx = \frac{1}{a}\sec^{-1}\left(\frac{|x|}{a}\right) + C$$

Note: More Anti-derivative formulas can be deduced from the reverse direction of the derivatives.

Table of Laplace Transforms

Let $u(t)$ be the unit step function, a, b, c, α, β be real numbers and $n = 1, 2, 3, ...$

	$f(t)$	$F(s) = \mathscr{L}\{f(t)\}$
L1.	1	$\frac{1}{s},\ s > 0$
L2.	e^{at}	$\frac{1}{s-a},\ s > a$
L3.	t^n	$\frac{n!}{s^{n+1}},\ s > 0$
L4.	$\sin(bt)$	$\frac{b}{s^2+b^2},\ s > 0$
L5.	$\cos(bt)$	$\frac{s}{s^2+b^2},\ s > 0$
L6.	$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}},\ s > a$
L7.	$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2},\ s > a$
L8.	$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2},\ s > a$
L9.	$\alpha f(t) + \beta g(t)$	$\alpha F(s) + \beta G(s)$
L10.	$e^{at} f(t)$	$F(s-a)$
L11.	$f'(t)$	$sF(s) - f(0)$
L12.	$f''(t)$	$s^2F(s) - sf(0) - f'(0)$
L13.	$f^{(n)}(t)$	$s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$
L14.	$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n}\big(F(s)\big) = (-1)^n F^{(n)}(s)$
L15.	$u_c(t) = u(t-c)$	$\frac{e^{-cs}}{s}$
L16.	$f(t) \cdot u_c(t)$	$e^{-cs}\mathscr{L}\{f(t+c)\}$
L17.	$f(t-c) \cdot u_c(t)$	$e^{-cs}F(s)$